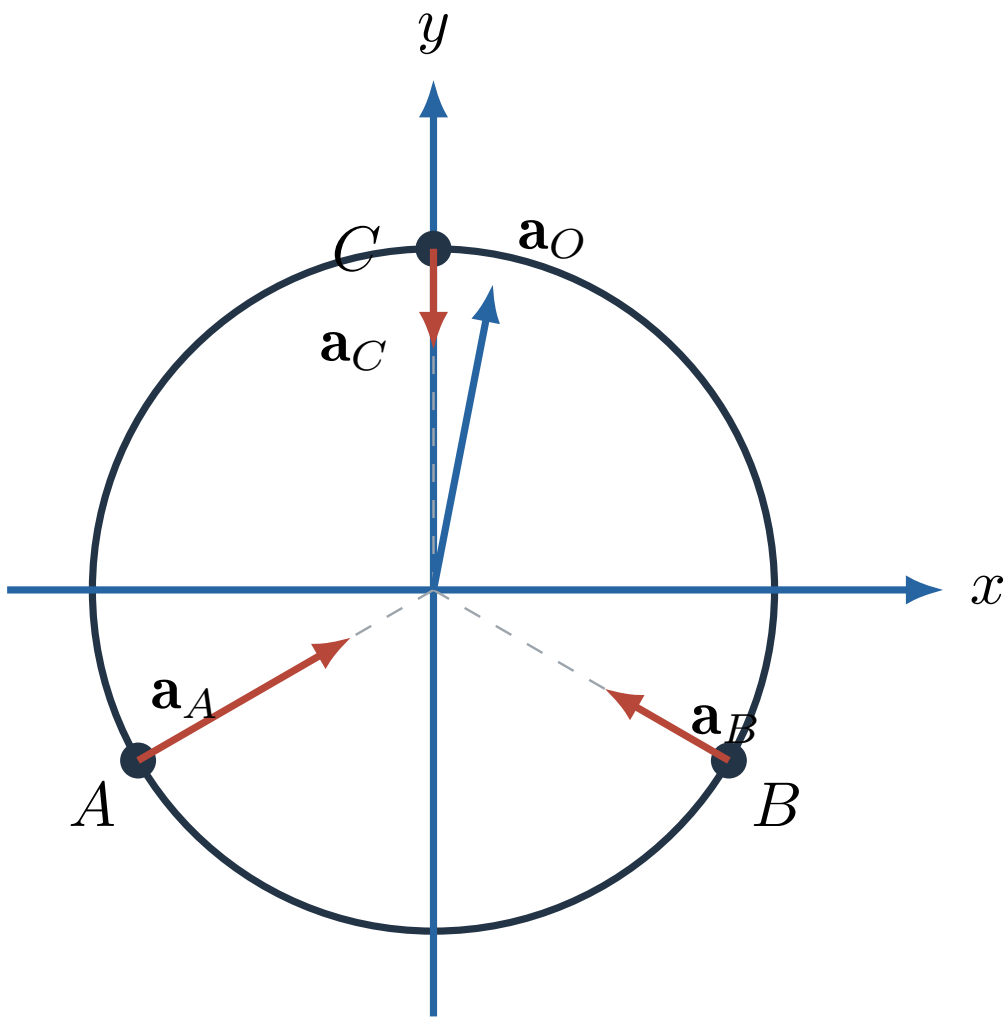

$$\mathbf{a}_O = (a_x, a_y)$$

$$a_y/2 + \sqrt{3}a_x/2 = a_A$$

$$a_y/2 - \sqrt{3}a_x/2 = a_B$$

$$a_C + a_y = v_0^2/R$$

红色箭头沿各小环受力方向；蓝色箭头为圆心加速度